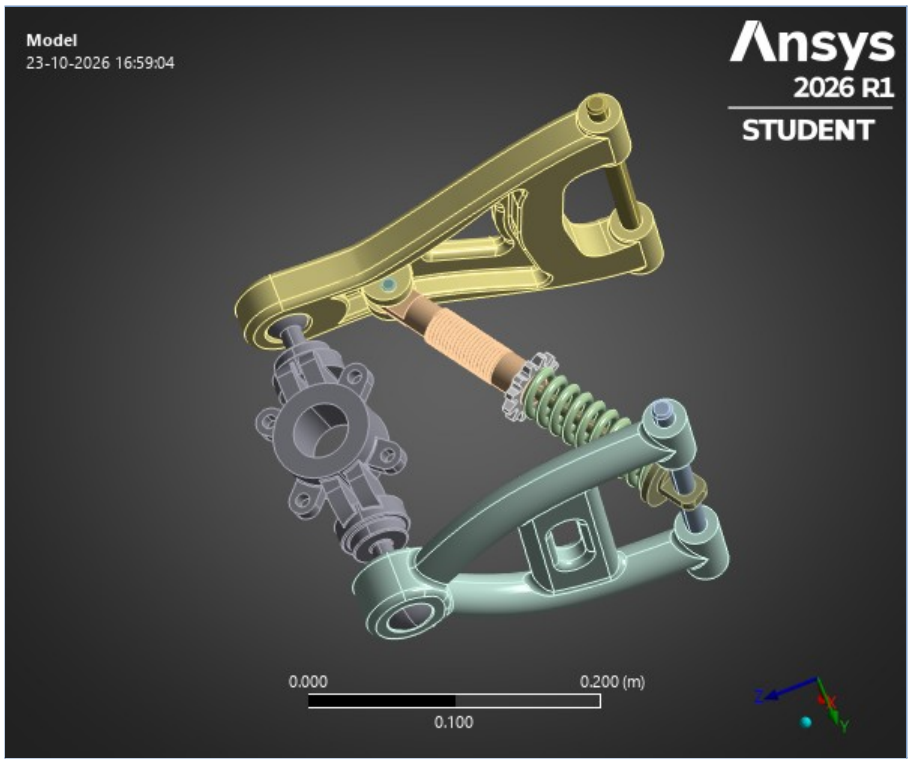
Project*

First Saved	Tuesday, July 2, 2024
Last Saved	Friday, October 23, 2026
Product Version	2026 R1
Imported Source	C:\Users\harsh\Downloads\SUSPENSION.mechdb
Imported Version	2026 R1
Save Project Before Solution	No
Save Project After Solution	No



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- **Material Data**
  - Aluminum Alloy
  - Chrome silicon

Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4, B4, C4)

TABLE 2
Model (A4, B4, C4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4, B4, C4) > Geometry Imports > Geometry Import (A3, B3, C3)

Object Name	*Geometry Import (A3, B3, C3)*
State	Solved
Definition	

Source	C:\Users\harsh\Downloads\suspension_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Advanced Geometry Options	
Compare Parts On Update	No
Analysis Type	3-D

Geometry

TABLE 4
Model (A4, B4, C4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	C:\Users\harsh\Downloads\suspension_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	0.3 m
Length Y	0.30031 m
Length Z	0.52073 m
Properties	
Volume	1.6537e-003 m³
Mass	4.7738 kg
Scale Factor Value	1.
Statistics	
Bodies	11
Active Bodies	10
Nodes	114542
Elements	68140
Mesh Metric	Skewness
Min	2.2233e-003
Max	0.99924
Average	0.40443
Standard Deviation	0.21749
Update Options	
Assign Default Material	No
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	Non-Associatively
Compare Parts Tolerance	Tight
Analysis Type	3-D
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4, B4, C4) > Geometry > Parts

Object Name	*Body*	*Mini_pin*	*pin1*	*Wheel connection*	*base*	*nut new*	*pin1*	*Spring*	*top*	*Upper_Arm*	*Lower_Arm*

				part							
State	Suppressed	Meshed									
Graphics Properties											
Visible	No	Yes									
Transparency	1										
Definition											
Suppressed	Yes	No									
Stiffness Behavior	Flexible										
Coordinate System	Default Coordinate System										
Reference Temperature	By Environment										
Treatment	None										
Material											
Assignment		Aluminum Alloy						Chrome silicon	Aluminum Alloy		
Nonlinear Effects	Yes										
Thermal Strain Effects	Yes										
Bounding Box											
Length X	0.3 m	3.5e-002 m	0.24 m	0.11193 m	4.5e-002 m	5.1749e-002 m	0.24 m	3.725e-002 m	2.7001e-002 m	0.21 m	
Length Y	0.3 m	1.e-002 m	1.5e-002 m	0.28501 m	0.16869 m	3.161e-002 m	1.5e-002 m	0.11517 m	0.129 m	4.1274e-002 m	4.7121e-002 m
Length Z	0.27 m	1.e-002 m	1.5e-002 m	0.11187 m	0.10921 m	4.7466e-002 m	1.5e-002 m	8.2116e-002 m	8.9387e-002 m	0.25281 m	0.29265 m
Properties											
Volume	1.8485e-002 m³	2.7196e-006 m³	4.224e-005 m³	3.5776e-004 m³	2.6824e-005 m³	8.1655e-006 m³	4.224e-005 m³	3.8365e-005 m³	5.5788e-005 m³	4.7614e-004 m³	6.0341e-004 m³
Centroid X	9.5071e-003 m	9.5072e-003 m		1.2132e-003 m	9.5071e-003 m	9.5408e-003 m	9.5072e-003 m	9.369e-003 m	9.5123e-003 m	9.2429e-003 m	9.5133e-003 m
Centroid Y	3.9749e-002 m	-7.2331e-002 m	-8.8106e-002 m	3.6602e-002 m	0.10715 m	3.2356e-002 m	0.15189 m	8.4721e-002 m	-1.3806e-002 m	0.14937 m	-9.1293e-002 m
Centroid Z	4.6566e-002 m	0.33228 m	0.1584 m	0.40778 m	0.22511 m	0.26976 m	0.1984 m	0.2386 m	0.29733 m	0.29725 m	0.2628 m
Moment of Inertia Ip1		9.2502e-008 kg·m²	3.2501e-006 kg·m²	5.6545e-003 kg·m²	2.092e-004 kg·m²	4.0127e-006 kg·m²	3.2501e-006 kg·m²	3.4882e-004 kg·m²	2.1843e-004 kg·m²	5.2216e-003 kg·m²	9.5256e-003 kg·m²
Moment of Inertia Ip2		7.9601e-007 kg·m²	5.5608e-004 kg·m²	4.3576e-004 kg·m²	7.9096e-006 kg·m²	7.7557e-006 kg·m²	5.5608e-004 kg·m²	6.9791e-005 kg·m²	1.3861e-005 kg·m²	9.6023e-003 kg·m²	1.356e-002 kg·m²
Moment of Inertia Ip3		7.9601e-007 kg·m²	5.5608e-004 kg·m²	5.6757e-003 kg·m²	2.0867e-004 kg·m²	4.011e-006 kg·m²	5.5608e-004 kg·m²	3.4866e-004 kg·m²	2.1775e-004 kg·m²	4.5473e-003 kg·m²	4.2938e-003 kg·m²
Mass		7.5332e-003 kg	0.117 kg	0.991 kg	7.4302e-002 kg	2.2618e-002 kg	0.117 kg	0.29946 kg	0.15453 kg	1.3189 kg	1.6714 kg
Statistics											
Nodes	0	200	499	30722	1398	8101	499	1948	21645	13460	36070
Elements	0	24	63	18922	661	4736	63	733	12742	7862	22334
Mesh Metric	Skewness										
Min	0.	0.33333		2.2233e-003	6.8719e-003	1.8904e-002	0.33335	7.079e-002	1.3491e-002	3.3718e-003	2.5668e-003
Max	0.	0.5065		0.97468	0.94365	0.99924	0.5065	0.98517	0.98863	0.98208	0.98686
Average	0.	0.37694	0.37064	0.36516	0.32212	0.59773	0.35019	0.61689	0.56973	0.32794	0.32507
Standard Deviation	0.	7.6408e-002	5.2404e-002	0.19054	0.19825	0.23731	5.1121e-002	0.19744	0.21013	0.18087	0.16609

TABLE 6
Model (A4, B4, C4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	5
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4, B4, C4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined

Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Remote Points

TABLE 8
Model (A4, B4, C4) > Remote Points

Object Name	*Remote Points*
State	Fully Defined

TABLE 9
Model (A4, B4, C4) > Remote Points > Remote Point

Object Name	*Remote Point*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	2 Faces
Coordinate System	Global Coordinate System
X Coordinate	-7.2514e-003 m
Y Coordinate	2.6557e-002 m
Z Coordinate	0.42282 m
Location	Defined
Definition	
Suppressed	No
Behavior	Deformable
Formulation	MPC
Relaxation Method	No
Pinball Region	All
DOF Selection	Program Controlled
Pilot Node APDL Name	

Connections

TABLE 10
Model (A4, B4, C4) > Connections

Object Name	*Connections*
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	0
Active Contacts	0
Joints	14
Active Joints	14
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 11
Model (A4, B4, C4) > Connections > Joints

Object Name	*Joints*	*Joints 2*

State	Fully Defined	
Definition		
Connection Type	Joint	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Auto Detection		
Tolerance Type	Slider	
Tolerance Slider	0.	
Tolerance Value	1.6796e-003 m	
Use Range	No	
Face-Face Angle Tolerance	75. °	
Group By	Bodies	
Search Across	Parts	
Fixed Joints	No	
Revolute Joints	No	
Statistics		
Connections	12	2
Active Connections	12	2

TABLE 12
Model (A4, B4, C4) > Connections > Joints > Joints

Object Name	*Revolute - Mini_pin To top*	*Spherical - Lower_Arm To Wheel connection part*	*Spherical - Upper_Arm To Wheel connection part*	*Translational - Ground To Wheel connection part*	*Fixed - Ground To pin1*	*Fixed - Ground To pin1*	*Translational - top To base*	*Fixed - nut new To top*	*Revolute - pin1 To Upper_Arm*	*Revolute - pin1 To Lower_Arm*	*Fixed - Lower_Arm To Mini_pin*
State	Fully Defined										
Definition											
Connection Type	Body-Body			Body-Ground			Body-Body				
Type	Revolute	Spherical		Translational	Fixed		Translational	Fixed	Revolute		Fixed
Torsional Stiffness	0. N·m/°								0. N·m/°		
Torsional Damping	0. N·m·s/°								0. N·m·s/°		
Element APDL Name											
Suppressed	No										
Friction Coefficient		0.									
Radius		0. m									
Solver Element Type					Program Controlled			Program Controlled			Program Controlled
Reference											
Scoping Method	Geometry Selection						Geometry Selection				
Applied By	Remote Attachment						Remote Attachment				
Scope	2 Faces						1 Face		4 Edges		
Body	Mini_pin	Lower_Arm	Upper_Arm				top	nut new	pin1		Lower_Arm
Coordinate System	Reference Coordinate System										
Behavior	Rigid										
Formulation	MPC										
Relaxation Method	No										
Pinball Region	All						All				
Mobile											
Scoping Method	Geometry Selection										
Applied By	Remote Attachment										
Scope	2 Faces						1 Face		4 Edges		
Body	top	Wheel connection part			pin1		base	top	Upper_Arm	Lower_Arm	Mini_pin
Initial Position	Unchanged										
Behavior	Rigid										
Formulation	MPC										
Relaxation Method	No										
Pinball											

Region	All					
Stops						
RZ Min Type	None				None	
RZ Max Type	None				None	
RX Min Type		None				
RX Max Type		None				
RY Min Type		None				
RY Max Type		None				
X Min Type			None		None	
X Max Type			None		None	

TABLE 13
Model (A4, B4, C4) > Connections > Joints > Joints

Object Name	*Revolute - pin1 To base*
State	Fully Defined
Definition	
Connection Type	Body-Body
Type	Revolute
Torsional Stiffness	0. N·m/°
Torsional Damping	0. N·m·s/°
Element APDL Name	
Suppressed	No
Reference	
Scoping Method	Geometry Selection
Applied By	Remote Attachment
Scope	2 Faces
Body	pin1
Coordinate System	Reference Coordinate System
Behavior	Rigid
Formulation	MPC
Relaxation Method	No
Pinball Region	All
Mobile	
Scoping Method	Geometry Selection
Applied By	Remote Attachment
Scope	4 Edges
Body	base
Initial Position	Unchanged
Behavior	Rigid
Formulation	MPC
Relaxation Method	No
Pinball Region	All
Stops	
RZ Min Type	None
RZ Max Type	None

TABLE 14
Model (A4, B4, C4) > Connections > Joints 2 > Joints

Object Name	*Fixed - nut new To Spring*	*Fixed - base To Spring*
State	Fully Defined	
Definition		
Connection Type	Body-Body	
Type	Fixed	
Solver Element Type	Program Controlled	
Element APDL Name		
Suppressed	No	
Reference		
Scoping Method	Geometry Selection	
Applied By	Remote Attachment	
Scope	1 Face	
Body	nut new	base
Coordinate System	Reference Coordinate System	
Behavior	Rigid	
Formulation	MPC	
Relaxation Method	No	

Pinball Region	All
Mobile	
Scoping Method	Geometry Selection
Applied By	Remote Attachment
Scope	1 Face
Body	Spring
Initial Position	Unchanged
Behavior	Rigid
Formulation	MPC
Relaxation Method	No
Pinball Region	All

Mesh

TABLE 15
Model (A4, B4, C4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	1.5e-002 m
Sizing	
Mesh Sizing	Advanced
Mesh Defaults	Medium
Growth Rate	1.85
Max Size	Default (3.e-002 m)
Mesh Defeaturing	Yes
Defeature Size	1.2252e-004 m
Capture Curvature	Yes
Curvature Min Size	3.e-003 m
Curvature Normal Angle	70.395°
Capture Proximity	No
Bounding Box Diagonal	0.67182 m
Average Surface Area	1.2456e-003 m²
Minimum Edge Length	2.2524e-005 m
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (0.1)
Smoothing	High
Mesh Optimization	No
Mesh Metric	Skewness
Min	2.2233e-003
Max	0.99924
Average	0.40443
Standard Deviation	0.21749
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Default (2.7e-003 m)
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Quad Dominant
Sweepable Body Method	Sweep

Statistics	
Nodes	114542
Elements	68140
Show Detailed Statistics	No

TABLE 16
Model (A4, B4, C4) > Mesh > Mesh Controls

Object Name	*Face Sizing*	*Face Sizing 2*	*Face Sizing 3*
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	22 Faces	2 Faces	
Definition			
Suppressed	No		
Type	Element Size		
Element Size	1.8e-002 m		
Advanced			
Defeature Size	Default (1.2252e-004 m)		
Influence Volume	No		
Behavior	Soft		
Growth Rate	Default (1.85)		
Capture Curvature	No		
Capture Proximity	No		

Static Structural (A5)

TABLE 17
Model (A4, B4, C4) > Analysis

Object Name	*Static Structural (A5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 18
Model (A4, B4, C4) > Static Structural (A5) > Analysis Settings

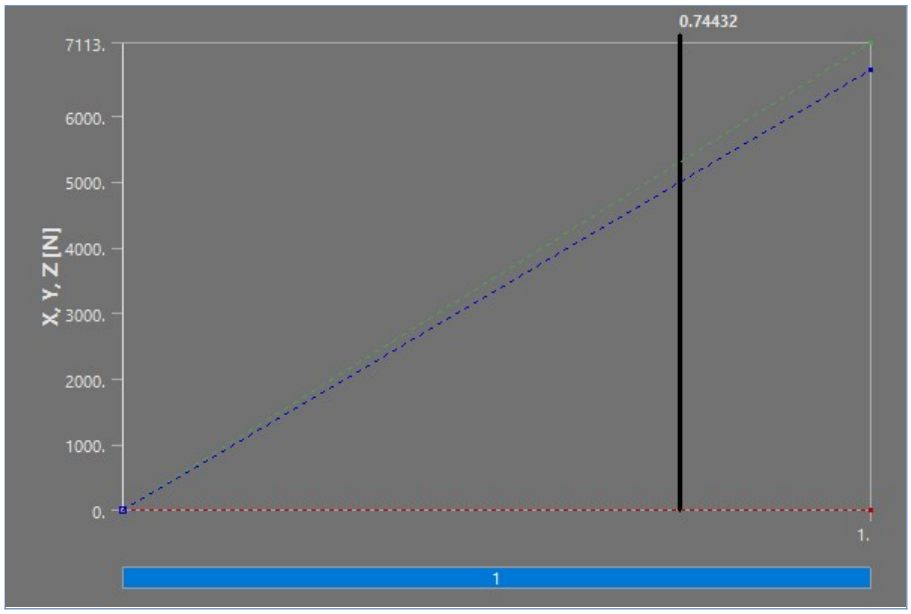
Object Name	*Analysis Settings*
State	Fully Defined
Restart Analysis	
Restart Type	Program Controlled
Status	Done
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Maximum Iterations Per Substep	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	Yes
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled

Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Off
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Options	
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\harsh\Downloads\suspension_files\dp0\SYS\MECH\
Future Analysis	Prestressed analysis
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	mks

TABLE 19
Model (A4, B4, C4) > Static Structural (A5) > Loads

Object Name	*Fixed Support*	*Remote Force*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	Remote Point
Geometry	3 Faces	
Remote Points		Remote Point
Coordinate System		Global Coordinate System
X Coordinate		-7.2514e-003 m
Y Coordinate		2.6557e-002 m
Z Coordinate		0.42282 m
Location		Defined
Definition		
Type	Fixed Support	Remote Force
Suppressed	No	
Define by		Components
X Component		0. N (ramped)
Y Component		7113. N (ramped)
Z Component		6690. N (ramped)
Behavior		Deformable
Advanced		
Pinball Region		All

FIGURE 1
Model (A4, B4, C4) > Static Structural (A5) > Remote Force



Solution (A6)

TABLE 20
Model (A4, B4, C4) > Static Structural (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	6 m 14 s
MAPDL Memory Used	2.8389 GB
MAPDL Result File Size	227.44 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 21
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 22
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Results

Object Name	*Total Deformation*	*Equivalent Stress Spring*	*Equivalent Stress Upper Arm*	*Equivalent Stress Lower Arm*	*Equivalent Stress Base Cylinder*	*Equivalent Stress Top Pin*	*Equivalent Stress Mini Pin*	*Equivalent Stress Knuckel*
State	Solved							
Scope								
Scoping Method	Geometry Selection							

Geometry	All Bodies	1 Body						1 Face
Definition								
Type	Total Deformation	Equivalent (von-Mises) Stress						
By	Time							
Display Time	Last							
Separate Data by Entity	No							
Calculate Time History	Yes							
Identifier								
Suppressed	No							
Results								
Minimum	0. m	4.519e+005 Pa	2213.3 Pa	43221 Pa	28587 Pa	0.49964 Pa	1.6984 Pa	1.2504e+007 Pa
Maximum	4.7339e-002 m	1.8099e+009 Pa	1.9475e+008 Pa	1.9549e+008 Pa	2.1931e+007 Pa	4.4102e+007 Pa	16.237 Pa	5.3182e+008 Pa
Average	2.7349e-002 m	7.7508e+008 Pa	1.1386e+007 Pa	9.3897e+006 Pa	1.6351e+006 Pa	2.1266e+006 Pa	5.9563 Pa	1.1615e+008 Pa
Minimum Occurs On	pin1	Spring	Upper_Arm	Lower_Arm	top	base	Mini_pin	Wheel connection part
Maximum Occurs On	Upper_Arm	Spring	Upper_Arm	Lower_Arm	top	base	Mini_pin	Wheel connection part
Minimum Value Over Time								
Minimum	0. m	2.2777e+005 Pa	2213.3 Pa	9395.3 Pa	21420 Pa	0.49964 Pa	1.6984 Pa	2.339e+006 Pa
Maximum	0. m	4.519e+005 Pa	10588 Pa	43221 Pa	28587 Pa	101.3 Pa	2275. Pa	1.2504e+007 Pa
Maximum Value Over Time								
Minimum	2.465e-002 m	9.3192e+008 Pa	3.6951e+007 Pa	4.6624e+007 Pa	1.1406e+007 Pa	2.234e+007 Pa	16.237 Pa	1.0253e+008 Pa
Maximum	4.7339e-002 m	1.8099e+009 Pa	1.9475e+008 Pa	1.9549e+008 Pa	2.1931e+007 Pa	4.4102e+007 Pa	18255 Pa	5.3182e+008 Pa
Information								
Time	1. s							
Load Step	1							
Substep	7							
Iteration Number	36							
Integration Point Results								
Display Option		Averaged						
Average Across Bodies		No						

FIGURE 2
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Total Deformation

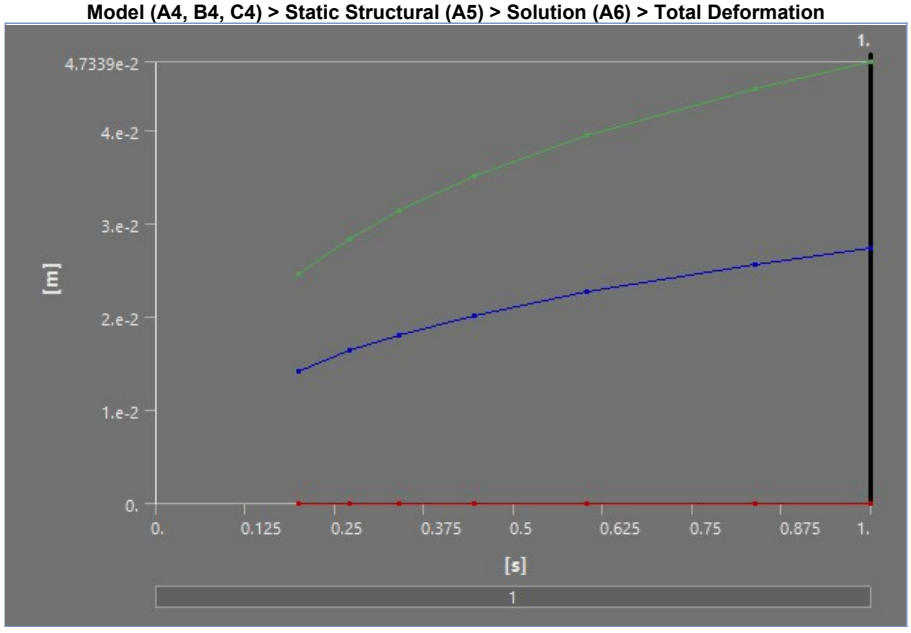


TABLE 23
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
0.2	0.	2.465e-002	1.4189e-002
0.27		2.8387e-002	1.6348e-002
0.34		3.1379e-002	1.8077e-002
0.445		3.5026e-002	2.0189e-002
0.6025		3.9382e-002	2.2715e-002
0.83875		4.4464e-002	2.5671e-002
1.		4.7339e-002	2.7349e-002

FIGURE 3
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Spring

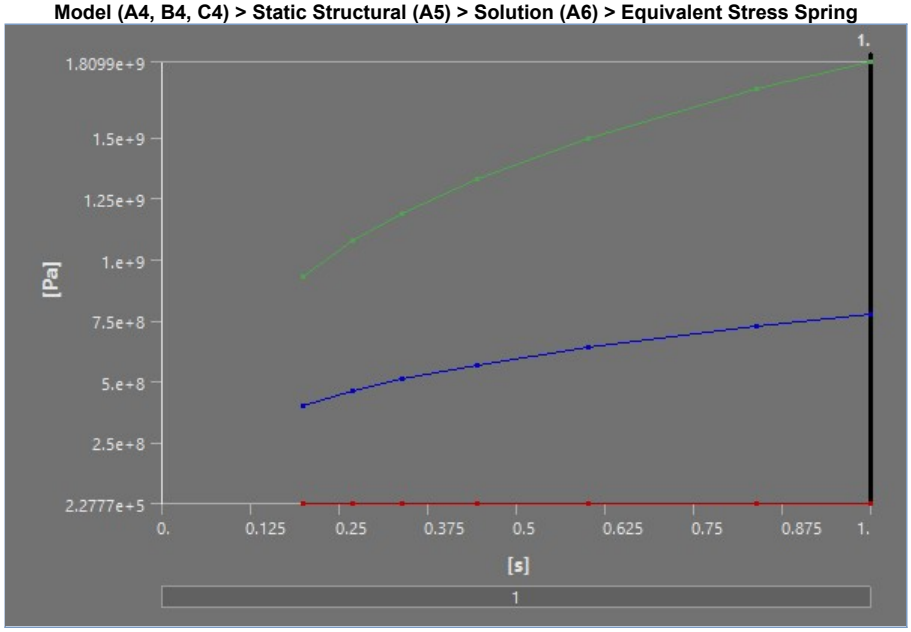


TABLE 24
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Spring

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
0.2	2.2777e+005	9.3192e+008	3.9759e+008
0.27	2.6376e+005	1.0749e+009	4.5886e+008
0.34	2.9266e+005	1.1898e+009	5.0814e+008
0.445	3.283e+005	1.3304e+009	5.6851e+008
0.6025	3.7119e+005	1.4991e+009	6.4106e+008
0.83875	4.2259e+005	1.6972e+009	7.2642e+008
1.	4.519e+005	1.8099e+009	7.7508e+008

FIGURE 4
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Upper Arm

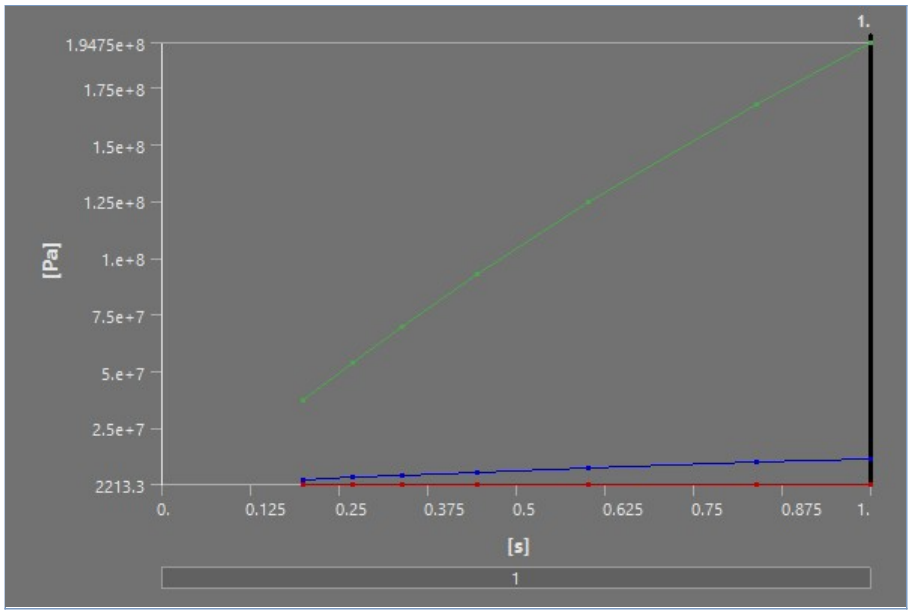


TABLE 25
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Upper Arm

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
0.2	5050.7	3.6951e+007	2.1992e+006
0.27	2658.7	5.366e+007	3.2362e+006
0.34	4568.7	6.9612e+007	4.2019e+006
0.445	6097.9	9.2792e+007	5.5416e+006
0.6025	10588	1.2464e+008	7.3765e+006
0.83875	5229.	1.6767e+008	9.8388e+006
1.	2213.3	1.9475e+008	1.1386e+007

FIGURE 5
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Lower Arm

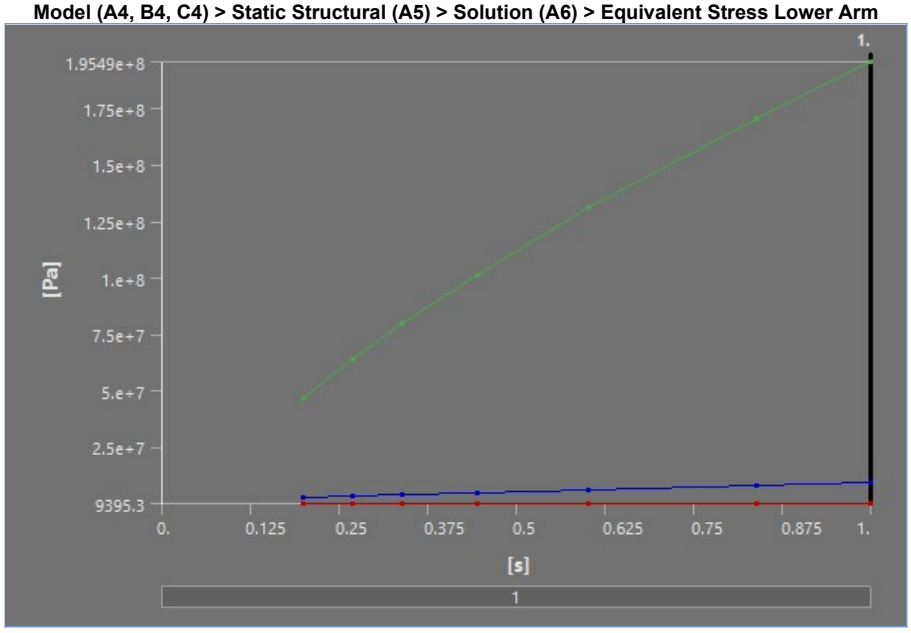


TABLE 26
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Lower Arm

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
0.2	9395.3	4.6624e+007	2.7065e+006
0.27	12955	6.377e+007	3.3596e+006
0.34	11672	7.9501e+007	3.993e+006
0.445	15864	1.0115e+008	4.9128e+006
0.6025	18769	1.3067e+008	6.2405e+006
0.83875	38265	1.7037e+008	8.1379e+006

1.	43221	1.9549e+008	9.3897e+006

FIGURE 6
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Base Cylinder

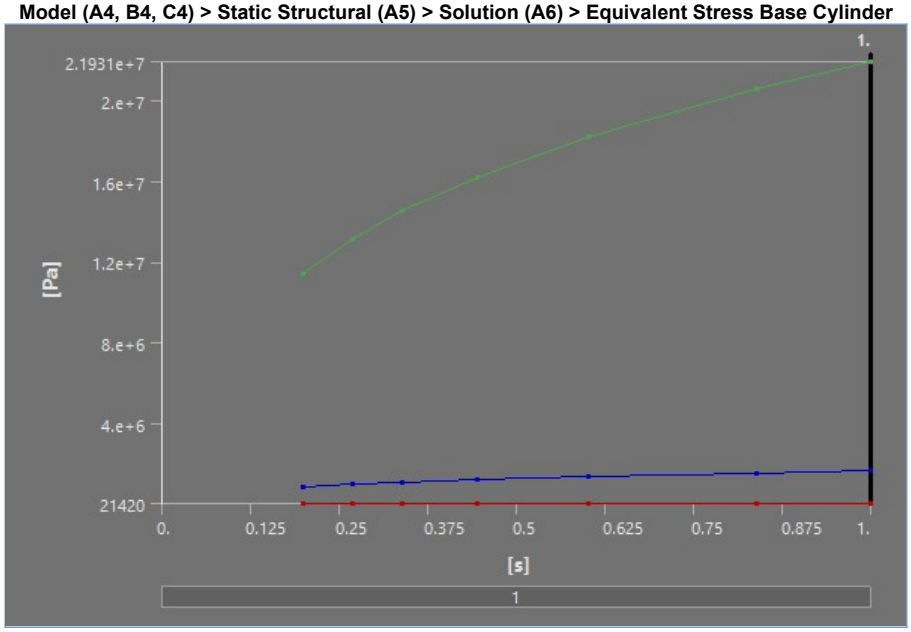


TABLE 27
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Base Cylinder

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
0.2	21420	1.1406e+007	8.3755e+005
0.27	22580	1.3138e+007	9.6683e+005
0.34	23366	1.4519e+007	1.0708e+006
0.445	24164	1.6207e+007	1.1982e+006
0.6025	25283	1.8219e+007	1.3514e+006
0.83875	26960	2.0591e+007	1.5321e+006
1.	28587	2.1931e+007	1.6351e+006

FIGURE 7
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Top Pin

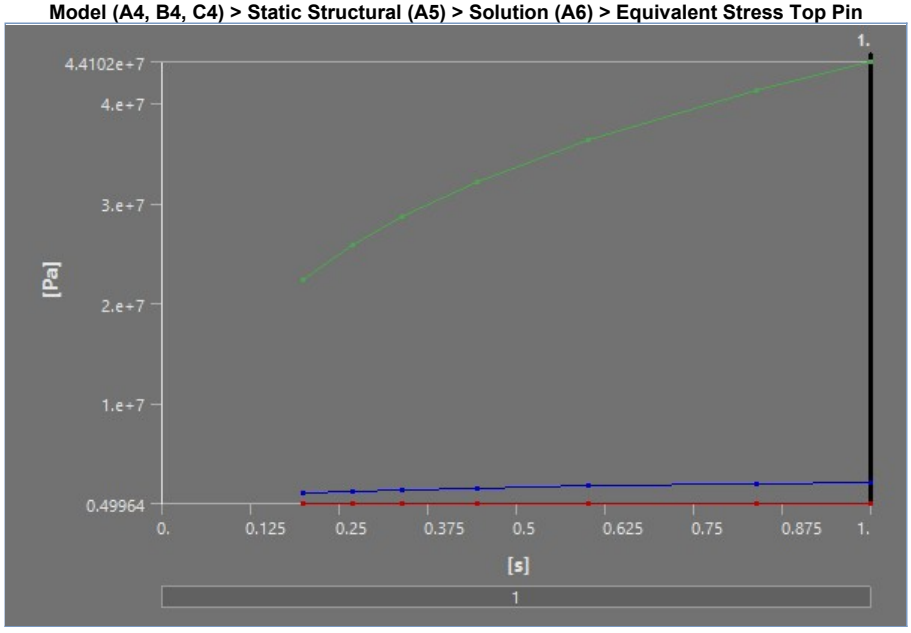


TABLE 28
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Top Pin

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
0.2	50.661	2.234e+007	1.0926e+006
0.27	7.3211	2.5829e+007	1.2605e+006

0.34	30.76	2.8645e+007	1.3956e+006
0.445	36.883	3.2109e+007	1.561e+006
0.6025	101.3	3.6294e+007	1.7598e+006
0.83875	6.8593	4.1256e+007	1.9934e+006
1.	0.49964	4.4102e+007	2.1266e+006

FIGURE 8
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Mini Pin

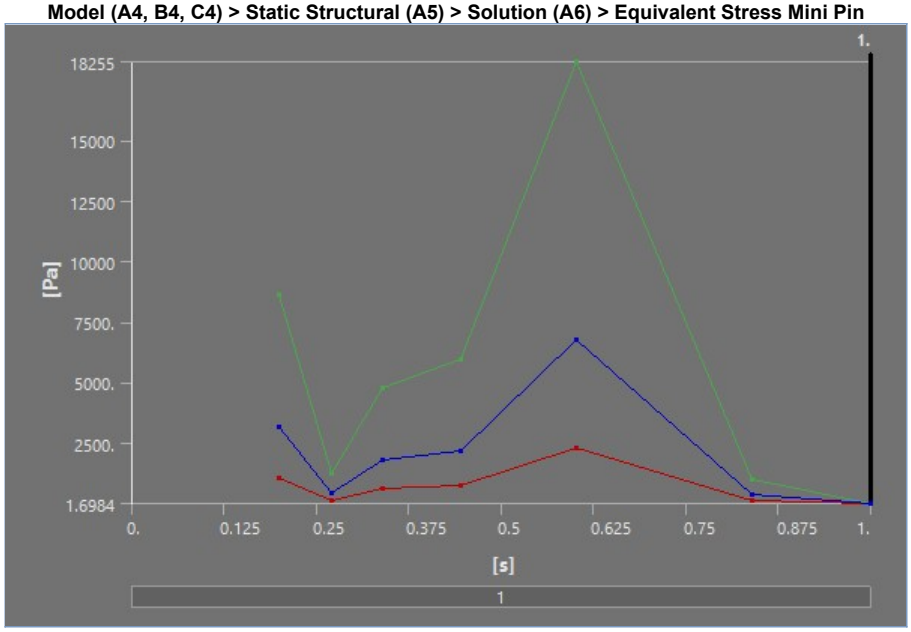


TABLE 29
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Mini Pin

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
0.2	1076.	8617.7	3194.3
0.27	153.99	1228.2	455.97
0.34	600.26	4810.3	1782.1
0.445	740.43	5936.7	2197.8
0.6025	2275.	18255	6758.4
0.83875	121.2	969.18	359.12
1.	1.6984	16.237	5.9563

FIGURE 9
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Knuckel

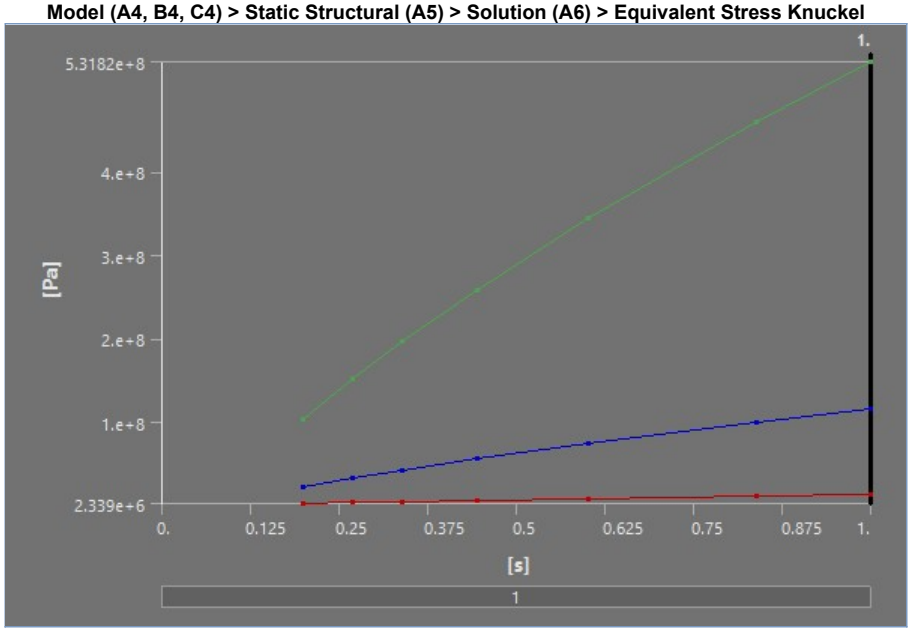


TABLE 30

Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress Knuckel

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
0.2	2.339e+006	1.0253e+008	2.2125e+007
0.27	3.4285e+006	1.5105e+008	3.2719e+007
0.34	4.4542e+006	1.962e+008	4.2581e+007
0.445	5.8978e+006	2.5884e+008	5.627e+007
0.6025	7.9144e+006	3.4456e+008	7.5029e+007
0.83875	1.0703e+007	4.5962e+008	1.0027e+008
1.	1.2504e+007	5.3182e+008	1.1615e+008

TABLE 31
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Stress Safety Tools

Object Name	*Stress Tool*
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Custom Value
Stress Limit	5.e+008 Pa

TABLE 32
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Stress Tool > Results

Object Name	*Safety Factor*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	2.5674
Minimum Occurs On	Upper_Arm
Minimum Value Over Time	
Minimum	2.5674
Maximum	13.531
Maximum Value Over Time	
Minimum	15.
Maximum	15.
Information	
Time	1. s
Load Step	1
Substep	7
Iteration Number	36

FIGURE 10
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

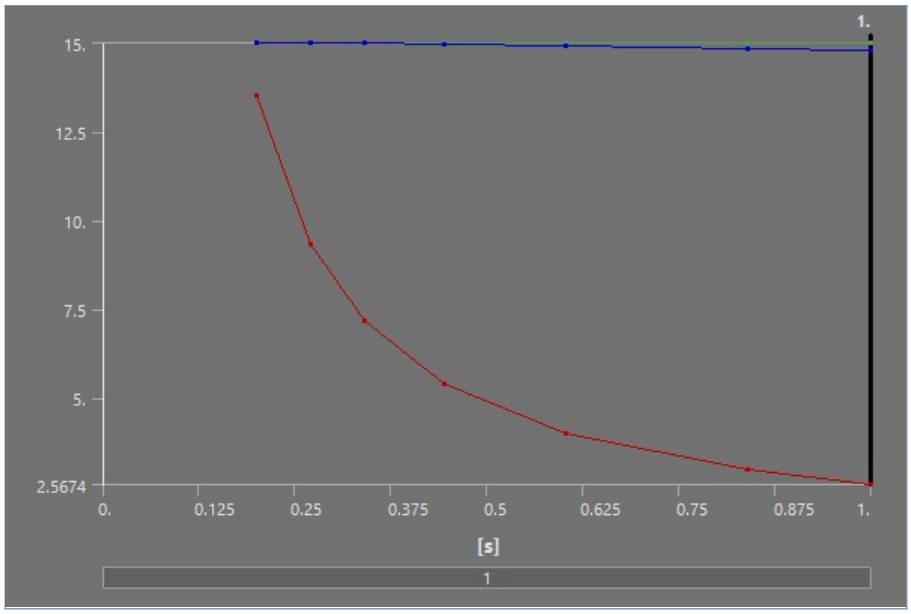


TABLE 33
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
0.2	13.531	15.	15.
0.27	9.318		14.996
0.34	7.1826		14.985
0.445	5.3884		14.958
0.6025	4.0116		14.908
0.83875	2.9821		14.83
1.	2.5674		14.771

TABLE 34
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Fatigue Tools

Object Name	*Fatigue Tool*
State	Solved
Domain	
Domain Type	Time
Materials	
Fatigue Strength Factor (Kf)	1.
Loading	
Type	Zero-Based
Scale Factor	1.
Definition	
Display Time	End Time
Options	
Analysis Type	Stress Life
Mean Stress Theory	None
Stress Component	Equivalent (von-Mises)
Life Units	
Units Name	cycles
1 cycle is equal to	1. cycles

FIGURE 11
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

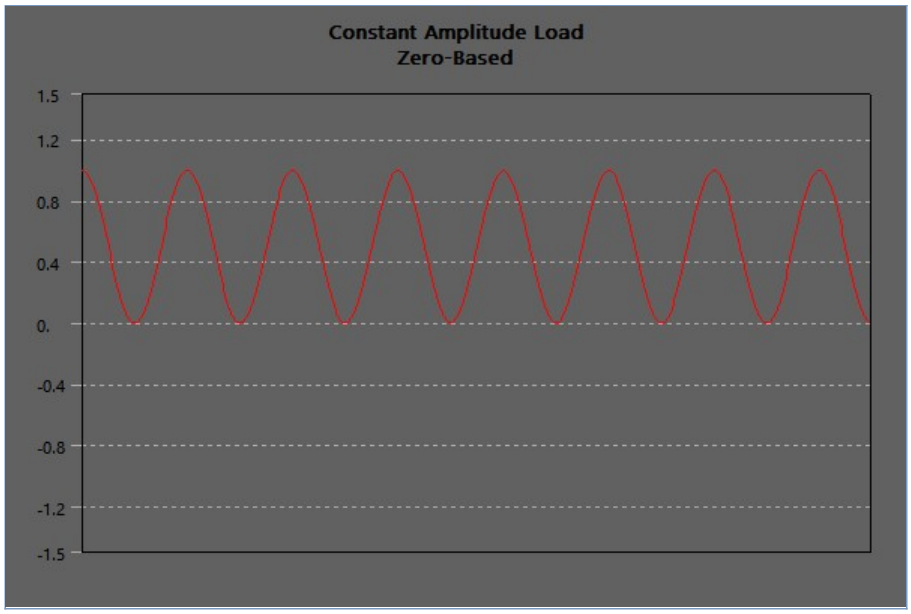


FIGURE 12
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

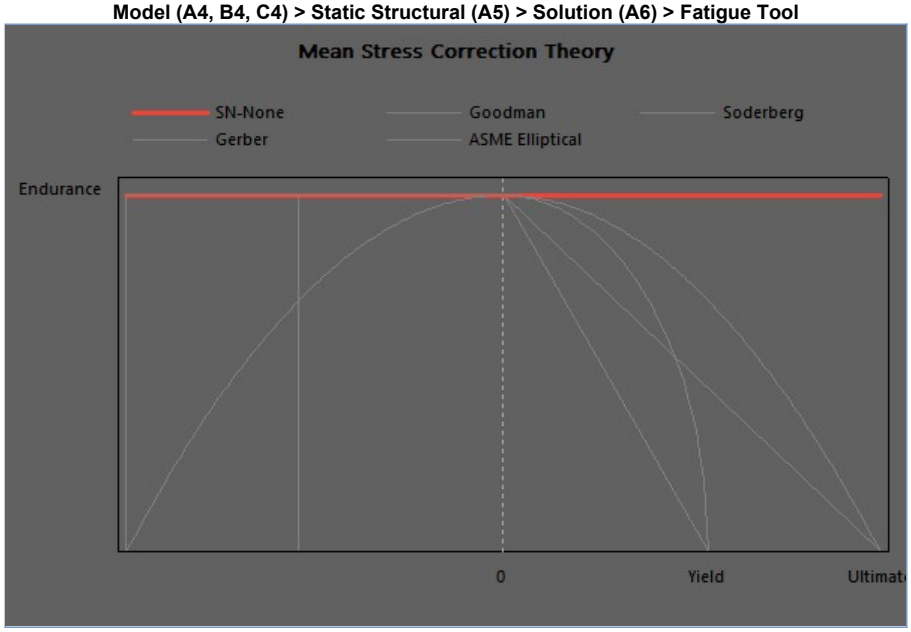


TABLE 35
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Results

Object Name	*Life*	*Damage*	*Safety Factor*
State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	2 Bodies		
Definition			
Type	Life	Damage	Safety Factor
Identifier			
Suppressed	No		
Design Life		1.e+009 cycles	
Results			
Minimum	2.188e+007 cycles		0.8465
Minimum Occurs On	Lower_Arm		Lower_Arm
Maximum		45.703	
Maximum Occurs On		Lower_Arm	

TABLE 36
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Probes

Object Name	*Force Reaction*
State	Solved
Definition	
Type	Force Reaction
Location Method	Boundary Condition
Boundary Condition	Fixed Support
Orientation	Global Coordinate System
Suppressed	No
Options	
Result Selection	All
Display Time	End Time
Results	
X Axis	895.29 N
Y Axis	-7070.4 N
Z Axis	-31410 N
Total	32208 N
Maximum Value Over Time	
X Axis	895.29 N
Y Axis	-1414.3 N
Z Axis	-6019.5 N
Total	32208 N
Minimum Value Over Time	
X Axis	367.73 N
Y Axis	-7070.4 N
Z Axis	-31410 N
Total	6194.3 N
Information	
Time	1. s
Load Step	1
Substep	7
Iteration Number	36

FIGURE 13
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Force Reaction

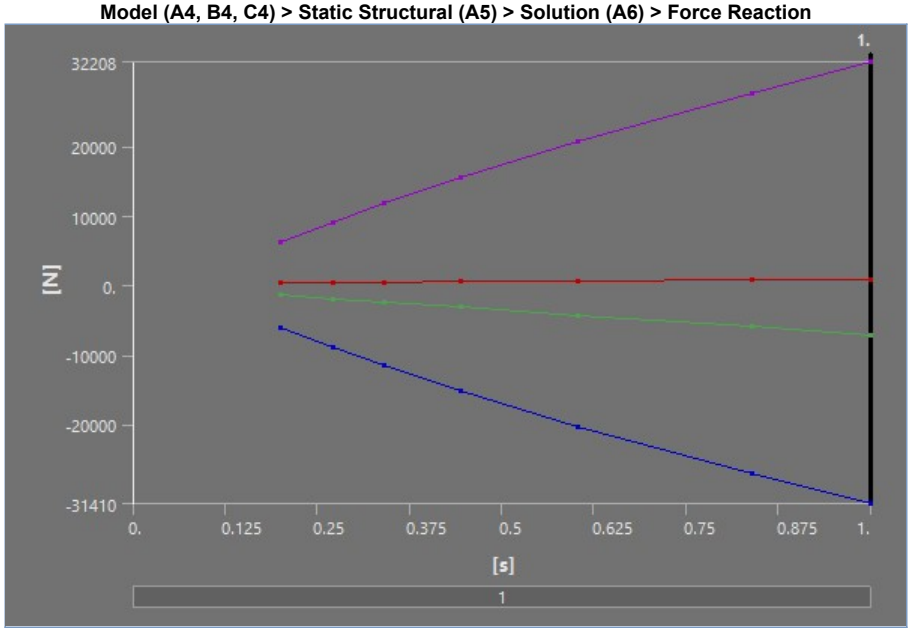


TABLE 37
Model (A4, B4, C4) > Static Structural (A5) > Solution (A6) > Force Reaction

Time [s]	Force Reaction (X) [N]	Force Reaction (Y) [N]	Force Reaction (Z) [N]	Force Reaction (Total) [N]
0.2	367.73	-1414.3	-6019.5	6194.3
0.27	442.74	-1908.1	-8881.5	9094.9
0.34	506.18	-2402.2	-11545	11803
0.445	587.52	-3144.	-15241	15573
0.6025	690.48	-4257.4	-20305	20758
0.83875	818.69	-5928.9	-27119	27772
1.	895.29	-7070.4	-31410	32208

Modal (B5)

TABLE 38
Model (A4, B4, C4) > Analysis

Object Name	*Modal (B5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Modal
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 39
Model (A4, B4, C4) > Modal (B5) > Initial Condition

Object Name	*Pre-Stress (Static Structural)*
State	Fully Defined
Definition	
Pre-Stress Environment	Static Structural
Pre-Stress Define By	Program Controlled
Reported Loadstep	Last
Reported Substep	Last
Reported Time	End Time
Contact Status	Use True Status
Newton-Raphson Option	Program Controlled
Load Control	Keep All Displacements As Zero

TABLE 40
Model (A4, B4, C4) > Modal (B5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Options	
Max Modes to Find	6
Limit Search to Range	No
Include Residual Vector	No
On Demand Expansion Option	Program Controlled
-- On Demand Expansion	No
Spin Softening	Program Controlled
Solver Controls	
Damped	No
Solver Type	Program Controlled
Rotordynamics Controls	
Coriolis Effect	Off
Campbell Diagram	Off
Output Controls	
Output Selection	None
Stress	No
Back Stress	No
Strain	No
Contact Data	No
Nodal Forces	No
Volume and Energy	No
Euler Angles	No
Calculate Reactions	No
General Miscellaneous	No
Result File Options	
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\harsh\Downloads\suspension_files\dp0\SYS-1\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	mks

Solution (B6)

TABLE 41

Model (A4, B4, C4) > Modal (B5) > Solution

Object Name	*Solution (B6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	57.33 s
MAPDL Memory Used	3.2812 GB
MAPDL Result File Size	38. MB
Post Processing	
Beam Section Results	No

The following bar chart indicates the frequency at each calculated mode.

FIGURE 14
Model (A4, B4, C4) > Modal (B5) > Solution (B6)

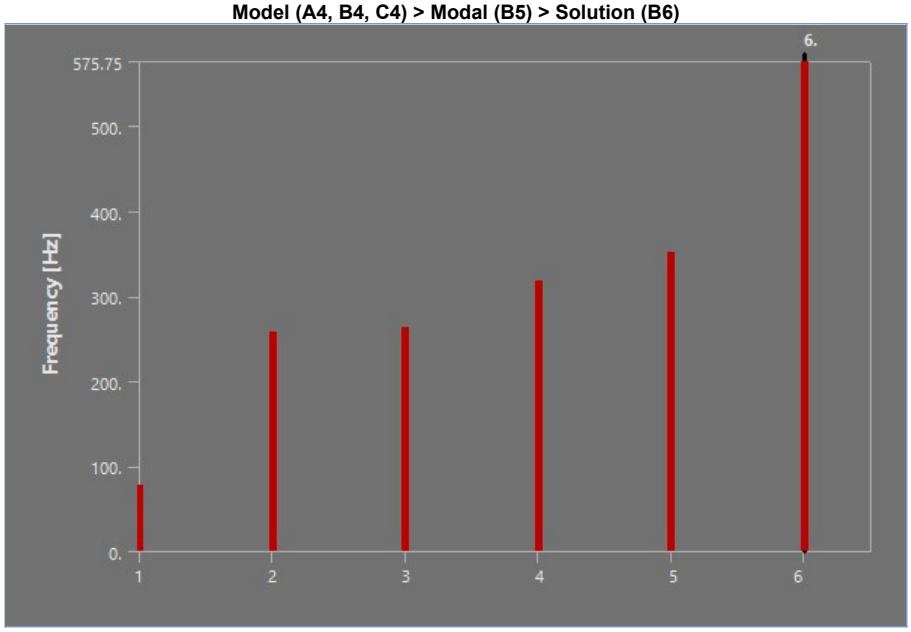


TABLE 42
Model (A4, B4, C4) > Modal (B5) > Solution (B6)

Mode	Frequency [Hz]
1.	77.38
2.	258.
3.	262.41
4.	318.26
5.	351.52
6.	575.75

TABLE 43
Model (A4, B4, C4) > Modal (B5) > Solution (B6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 44
Model (A4, B4, C4) > Modal (B5) > Solution (B6) > Results

Object Name	*Total Deformation*	*Total Deformation 2*	*Total Deformation 3*	*Total Deformation 4*	*Total Deformation 5*	*Total Deformation 6*
State	Solved					
Scope						
Scoping Method	Geometry Selection					
Geometry	All Bodies					
Definition						
Type	Total Deformation					
Mode	1.	2.	3.	4.	5.	6.
Separate Data by Entity	No					
Identifier						
Suppressed	No					
Results						
Minimum	0. m					
Maximum	0.80506 m	2.973 m	3.1165 m	2.9173 m	3.5127 m	3.2354 m
Average	0.47169 m	3.0641e-002 m	3.2263e-002 m	7.0069e-002 m	3.5166e-002 m	3.8796e-002 m
Minimum Occurs On	pin1					
Maximum Occurs On	Upper_Arm	Spring				
Information						
Frequency	77.38 Hz	258. Hz	262.41 Hz	318.26 Hz	351.52 Hz	575.75 Hz

TABLE 45
Model (A4, B4, C4) > Modal (B5) > Solution (B6) > Total Deformation

Mode	Frequency [Hz]
1.	77.38
2.	258.
3.	262.41
4.	318.26
5.	351.52
6.	575.75

TABLE 46
Model (A4, B4, C4) > Modal (B5) > Solution (B6) > Total Deformation 2

Mode	Frequency [Hz]
1.	77.38
2.	258.
3.	262.41
4.	318.26
5.	351.52
6.	575.75

TABLE 47
Model (A4, B4, C4) > Modal (B5) > Solution (B6) > Total Deformation 3

Mode	Frequency [Hz]
1.	77.38
2.	258.
3.	262.41
4.	318.26
5.	351.52
6.	575.75

TABLE 48
Model (A4, B4, C4) > Modal (B5) > Solution (B6) > Total Deformation 4

Mode	Frequency [Hz]
1.	77.38
2.	258.
3.	262.41
4.	318.26
5.	351.52
6.	575.75

TABLE 49
Model (A4, B4, C4) > Modal (B5) > Solution (B6) > Total Deformation 5

Mode	Frequency [Hz]
1.	77.38
2.	258.
3.	262.41
4.	318.26
5.	351.52

6.	575.75

TABLE 50
Model (A4, B4, C4) > Modal (B5) > Solution (B6) > Total Deformation 6

Mode	Frequency [Hz]
1.	77.38
2.	258.
3.	262.41
4.	318.26
5.	351.52
6.	575.75

Eigenvalue Buckling (C5)

TABLE 51
Model (A4, B4, C4) > Analysis

Object Name	*Eigenvalue Buckling (C5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Eigenvalue Buckling
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 52
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Initial Condition

Object Name	*Pre-Stress (Static Structural)*
State	Fully Defined
Definition	
Pre-Stress Environment	Nonlinear Static Structural
Pre-Stress Define By	Program Controlled
Reported Loadstep	Last
Reported Substep	Last
Reported Time	End Time
Contact Status	Use True Status

TABLE 53
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Options	
Max Modes to Find	2.
Keep Pre-Stress Load-Pattern	Yes
Solver Controls	
Solver Type	Program Controlled
Include Negative Load Multiplier	No
Output Controls	
Output Selection	None
Stress	No
Back Stress	No
Strain	No
Contact Data	No
Volume and Energy	No
Euler Angles	No
General Miscellaneous	No
Result File Options	
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\harsh\Downloads\suspension_files\dp0\SYS-2\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	mks

Solution (C6)

TABLE 54
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Solution

Object Name	*Solution (C6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	53.506 s
MAPDL Memory Used	4.0479 GB
MAPDL Result File Size	26.688 MB
Post Processing	
Beam Section Results	No

FIGURE 15
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Solution (C6)

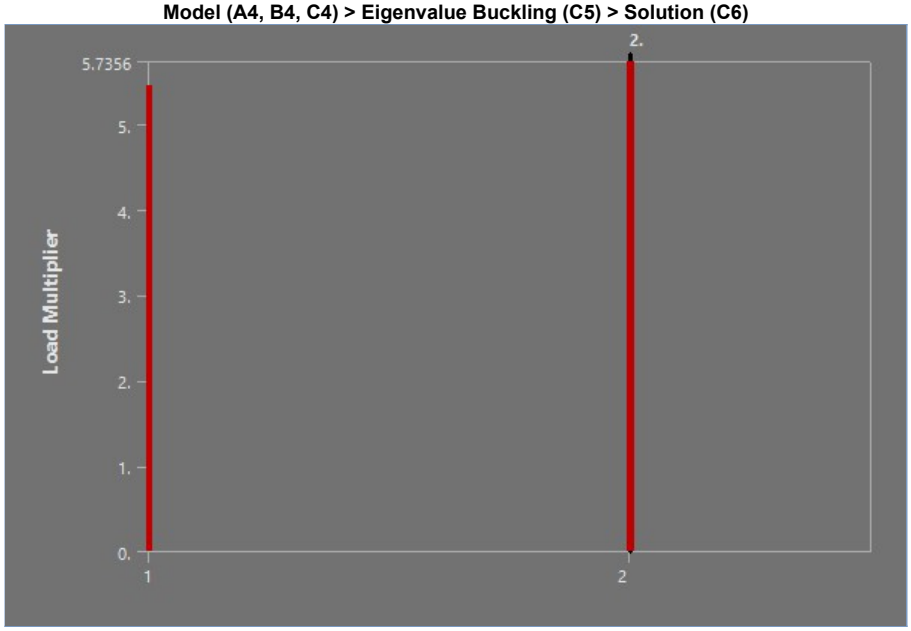


TABLE 55
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Solution (C6)

Mode	Load Multiplier
1.	5.462
2.	5.7356

TABLE 56
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Solution (C6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 57
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Solution (C6) > Results

Object Name	*Total Deformation*	*Total Deformation 2*
State	Solved	

Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Total Deformation	
Mode	1.	2.
Separate Data by Entity	No	
Identifier		
Suppressed	No	
Results		
Load Multiplier	5.462	5.7356
Minimum	0. m	
Maximum	1.0956 m	1.1815 m
Average	1.1109e-002 m	9.3619e-003 m
Minimum Occurs On	pin1	
Maximum Occurs On	Spring	

TABLE 58
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Solution (C6) > Total Deformation

Mode	Load Multiplier
1.	5.462
2.	5.7356

TABLE 59
Model (A4, B4, C4) > Eigenvalue Buckling (C5) > Solution (C6) > Total Deformation 2

Mode	Load Multiplier
1.	5.462
2.	5.7356

Material Data

Aluminum Alloy

TABLE 60
Aluminum Alloy > Constants

Density	2770 kg m^-3
Coefficient of Thermal Expansion	2.3e-005 C^-1
Specific Heat	875 J kg^-1 C^-1

TABLE 61
Aluminum Alloy > Color

Red	Green	Blue
138	104	46

TABLE 62
Aluminum Alloy > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
0

TABLE 63
Aluminum Alloy > Compressive Yield Strength

Compressive Yield Strength Pa
2.8e+008

TABLE 64
Aluminum Alloy > Tensile Yield Strength

Tensile Yield Strength Pa
2.8e+008

TABLE 65
Aluminum Alloy > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
3.1e+008

TABLE 66
Aluminum Alloy > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 67

Aluminum Alloy > Isotropic Thermal Conductivity

Thermal Conductivity W m^-1 C^-1	Temperature C
114	-100
144	0
165	100
175	200

TABLE 68
Aluminum Alloy > S-N Curve

Alternating Stress Pa	Cycles	R-Ratio
2.758e+008	1700	-1
2.413e+008	5000	-1
2.068e+008	34000	-1
1.724e+008	1.4e+005	-1
1.379e+008	8.e+005	-1
1.172e+008	2.4e+006	-1
8.963e+007	5.5e+007	-1
8.274e+007	1.e+008	-1
1.706e+008	50000	-0.5
1.396e+008	3.5e+005	-0.5
1.086e+008	3.7e+006	-0.5
8.791e+007	1.4e+007	-0.5
7.757e+007	5.e+007	-0.5
7.239e+007	1.e+008	-0.5
1.448e+008	50000	0
1.207e+008	1.9e+005	0
1.034e+008	1.3e+006	0
9.308e+007	4.4e+006	0
8.618e+007	1.2e+007	0
7.239e+007	1.e+008	0
7.412e+007	3.e+005	0.5
7.067e+007	1.5e+006	0.5
6.636e+007	1.2e+007	0.5
6.205e+007	1.e+008	0.5

TABLE 69
Aluminum Alloy > Isotropic Resistivity

Resistivity kg m^3 A^-2 s^-3	Temperature C
2.43e-008	0
2.67e-008	20
3.63e-008	100

TABLE 70
Aluminum Alloy > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
7.1e+010	0.33	6.9608e+010	2.6692e+010	

TABLE 71
Aluminum Alloy > Isotropic Relative Permeability

Relative Permeability
1

Chrome silicon

TABLE 72
Chrome silicon > Constants

Density	7805.7 kg m^-3

TABLE 73
Chrome silicon > Color

Red	Green	Blue
222	222	222

TABLE 74
Chrome silicon > Melting Temperature

Melting Temperature C
246.11

TABLE 75
Chrome silicon > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C

2.e+011	0.29	1.5873e+011	7.7519e+010	